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Sakai; Masayuki et al.

Magnetic geared electrical machine and power generation system using the same

Abstract

A magnetic geared electrical machine includes: a stator including a stator coil; a first rotor including a plurality of pole pieces; a second rotor including a plurality of rotor magnets, and arranged opposite to the stator across the first rotor in a radial direction; a housing having a first inlet opening and a second inlet opening, and supporting the stator; a first inlet cavity defined by at least the housing so as to communicate with the first inlet opening provided in the housing and with a first radial gap between the stator and the first rotor; and a second inlet cavity defined by at least the housing so as to communicate with the second inlet opening provided in the housing and with a second radial gap between the first rotor and the second rotor.

Inventors:	Sakai; Masayuki (Tokyo, JP), Aoki; Yasutaka (Tokyo, JP), Yuge; Atsushi (Tokyo, JP), Matsushita; Takatoshi (Tokyo, JP)
Applicant:	mitsubishi heavy industries, ltd. (Tokyo, JP)
Family ID:	82654557
Assignee:	mitsubishi heavy industries, ltd. (Tokyo, JP)
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Primary Examiner: Nguyen; Tran N

Attorney, Agent or Firm: Osha Bergman Watanabe & Burton LLP

Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a magnetic geared electrical machine and a power generation system using the same.

(2) This application claims the priority of Japanese Patent Application No. 2021-012757 filed on Jan. 29, 2021, the content of which is incorporated herein by reference.

BACKGROUND ART

(3) Conventionally, a magnetic geared electrical machine is known that combines a magnetic gear device for converting a rotation speed between a high-speed rotor and a low-speed rotor with a generator or an electric motor.

(4) For example, Patent Document 1 describes a magnetic geared electrical machine that comprises a first rotor including a support structure, a second rotor, and a stator, and is configured such that at least one of the second rotor and stator accommodates the support structure.

(5) Further, although not related to the magnetic geared electrical machine, Patent Document 2 describes a magnetic gear device in which a cooling medium flow path is disposed between electromagnetic steel magnetic path members (pole pieces) arranged in the circumferential direction.

CITATION LIST

Patent Literature

(6) Patent Document 1: U.S. Pat. No. 9,219,395B Patent Document 2: JP2012-246982A

SUMMARY

Problems to be Solved

(7) In the magnetic geared electrical machine described in Patent Document 1, for example, as a countermeasure against heat generation due to copper loss in the stator coils or iron loss in the pole pieces, each part of the magnetic geared electrical machine is cooled as appropriate.

(8) The amount of heat generated is not always equal on the stator-side region and the rotor-side region of the magnetic geared electrical machine, with the pole pieces as the boundary. Also, the amount of heat dissipated outside the system by heat conduction, heat transfer, and radiation is not always equal on the stator-side region and the rotor-side region of the magnetic geared electrical machine, with the pole pieces as the boundary.

(9) Therefore, cooling design according to the amount of heat generated or dissipated in each region on both sides of the pole pieces in the radial direction of the magnetic geared electrical machine is desirable.

(10) In the magnetic gear device described in Patent Document 2, cooling design that takes into account the amount of heat generated or dissipated in each region on both sides of the pole pieces is impossible because cooling medium flow paths are formed between the circumferentially arranged pole pieces.

(11) In view of the above, an object of some embodiments of the present invention is to provide a magnetic geared electrical machine that facilitates cooling design according to the amount of heat generated or dissipated in each region on both sides of the pole pieces in the radial direction.

Solution to the Problems

(12) A magnetic geared electrical machine according to at least one embodiment of the present invention includes: a stator including a stator coil; a first rotor including a plurality of pole pieces; a second rotor including a plurality of rotor magnets, and arranged opposite to the stator across the first rotor in a radial direction; a housing having a first inlet opening and a second inlet opening, and supporting the stator; a first inlet cavity defined by at least the housing so as to communicate with the first inlet opening provided in the housing and with a first radial gap between the stator

and the first rotor; and a second inlet cavity defined by at least the housing so as to communicate with the second inlet opening provided in the housing and with a second radial gap between the first rotor and the second rotor.

Advantageous Effects

(13) According to at least one embodiment of the present invention, it is possible to easily make cooling design according to the amount of heat generated or dissipated in each region on both sides of the pole pieces in the radial direction.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1A is a schematic view showing an example of a magnetic geared electrical machine.
- (2) FIG. 1B is a schematic view showing another example of the magnetic geared electrical machine.
- (3) FIG. 2 is a radial cross-sectional view of the magnetic geared electrical machine according to an embodiment.
- (4) FIG. 3 is a diagram showing the magnetic geared electrical machine according to an embodiment.
- (5) FIG. 4 is a diagram showing the magnetic geared electrical machine according to another embodiment.
- (6) FIG. 5A is a cross-sectional view of a partition member according to an embodiment.
- (7) FIG. 5B is a cross-sectional view of the partition member according to another embodiment.
- (8) FIG. 5C is a cross-sectional view of the partition member according to another embodiment.
- (9) FIG. 6 is a diagram showing the magnetic geared electrical machine according to an embodiment.
- (10) FIG. 7 is a diagram showing the magnetic geared electrical machine according to an embodiment.
- (11) FIG. 8A is a diagram showing an arrangement example of the partition member.
- (12) FIG. 8B is a diagram showing another arrangement example of the partition member.
- (13) FIG. 9 is a diagram showing another arrangement example of the partition member.

DETAILED DESCRIPTION

(14) Embodiments of the present invention will now be described in detail with reference to the accompanying drawings. It is intended, however, that unless particularly identified, dimensions, materials, shapes, relative positions, and the like of components described in the embodiments shall be interpreted as illustrative only and not intended to limit the scope of the present invention.

(15) (Overview of Magnetic Geared Electrical Machine)

(16) FIGS. 1A and 1B are each a schematic view showing an example of a magnetic geared electrical machine. Herein, in FIGS. 1A and 1B, the “axial direction” is a direction parallel to a rotational axis of a rotor (30, 40) of a magnetic geared electrical machine 10, and the “radial direction” is a direction orthogonal to the rotational axis of the rotor (30, 40).

(17) In an embodiment, as shown in FIG. 1A, the magnetic geared electrical machine 10 is a magnetic geared generator 10A configured to be driven by a prime mover 2 to generate power and supply electric power P generated by the power generation to a power supply destination 4 which may be, for example, a power grid.

(18) In another embodiment, as shown in FIG. 1B, the magnetic geared electrical machine 10 is a magnetic geared motor 10B configured to drive a rotating machine 8 upon receiving the electric power P supplied from a power supply source 6 which may be, for example, a power grid.

(19) In the embodiment shown in FIG. 1A, the magnetic geared generator 10A constitutes part of a power generation system 1. The power generation system 1 may be, for example, a renewable

energy power generation system such as a wind power generation system or a tidal current power generation system. If the power generation system **1** is a wind power generation system, the prime mover **2** is a wind turbine rotor.

(20) The magnetic geared generator **10A** includes a stator **20** with a stator magnet **22** and a stator coil **24**, a first rotor **30** with a pole piece **32**, and a second rotor **40** with a rotor magnet **42**. In the example shown in FIG. **1A**, the stator **20** is arranged inside a housing **21** for supporting a rotational shaft **3** of the prime mover **2** via a bearing **B1**. The first rotor **30** is configured to rotate with the rotational shaft **3** of the prime mover **2**. The first rotor **30** includes first end plates **34** respectively disposed at both axial ends of the pole piece **32**, and each of the first end plates **34** is connected to the rotational shaft **3**. The second rotor **40** includes second end plates **44** respectively disposed at the both axial ends of the rotor magnet **42**. Each of the second end plates **44** is mounted on the rotational shaft **3** (or the first rotor **30** rotating together with the rotational shaft **3**) via the bearing **B2** so as to allow the second rotor **40** to rotate faster than the rotational shaft **3** and the first rotor **30**. The second rotor **40** is disposed within the region surrounded by the first rotor **30** including the pole piece **32** and the first end plates **34** and the rotational shaft **3**.

(21) In the embodiment shown in FIG. **1A**, the magnetic geared generator **10A** has a configuration in which the stator **20**, the first rotor **30**, and the second rotor **40** are arranged in order radially inward. In another embodiment, the magnetic geared generator **10A** has a configuration in which the second rotor **40**, the first rotor **30**, and the stator **20** are arranged in order radially inward. In this case, the second rotor **40**, the first rotor **30**, and the stator **20** are arranged radially inward of the cylindrical rotational shaft **3**.

(22) The above-described magnetic geared generator **10A** is an integrated combination of a magnetic gear and a generator, and is configured to convert a mechanical input from the prime mover **2** into power by utilizing a harmonic type magnetic gear principle and electromagnetic induction.

(23) For example, power generation in the magnetic geared generator **10A** may be performed according to the following principle. A magnetic flux of the stator magnet **22** is modulated by the pole piece **32** of the first rotor (low-speed rotor) **30** rotating together with the rotational shaft **3** of the prime mover **2**, and the rotor magnet **42** receives a magnetic force from the modulated magnetic field, thereby rotating the second rotor (high-speed rotor) **40**. At this time, the ratio (speed increasing ratio) of the rotation speed of the second rotor (high-speed rotor) **40** to the first rotor (low-speed rotor) **30** is expressed by a ratio of the number of poles N_i of the pole piece **32** to the number of pole pairs N_H of the rotor magnet **42** ($=N_i/N_H$). When the second rotor (high-speed rotor) **40** rotates, an electric current is generated in the stator coil **24** by electromagnetic induction.

(24) In the embodiment shown in FIG. **1B**, the basic configuration of the magnetic geared motor **10B** is common to the magnetic geared generator **10A** shown in FIG. **1A**.

(25) That is, the magnetic geared motor **10B** includes the stator **20** with the stator magnet **22** and the stator coil **24**, the first rotor **30** with the pole piece **32**, and the second rotor **40** with the rotor magnet **42**. In the example shown in FIG. **1B**, the stator **20** is fixed to the inside of the housing **21** for supporting a rotational shaft **9** of the rotating machine **8** via the bearing **B1**. The first rotor **30** includes first end plates **34** respectively disposed at both axial ends of the pole piece **32**, and each of the first end plates **34** is connected to the rotational shaft **9**. The second rotor **40** includes second end plates **44** respectively disposed at the both axial ends of the rotor magnet **42**. Each of the second end plates **44** is mounted on the rotational shaft **9** (or the first rotor **30** rotating together with the rotational shaft **9**) via the bearing **B2** so as to allow the second rotor **40** to rotate faster than the rotational shaft **9** and the first rotor **30**. The second rotor **40** is disposed within the region surrounded by the first rotor **30** including the pole piece **32** and the first end plates **34** and the rotational shaft **9**.

(26) In the embodiment shown in FIG. **1B**, the magnetic geared motor **10B** has a configuration in which the stator **20**, the first rotor **30**, and the second rotor **40** are arranged in order radially inward.

In another embodiment, the magnetic geared motor **10B** has a configuration in which the second rotor **40**, the first rotor **30**, and the stator **20** are arranged in order radially inward. In this case, the second rotor **40**, the first rotor **30**, and the stator **20** are arranged radially inward of the cylindrical rotational shaft **9**.

(27) The magnetic geared motor **10B** is an integrated combination of a magnetic gear and a motor, rotates the second rotor (high-speed rotor) **40** by a rotating magnetic field generated by energizing the stator coil **24**, and utilizes the principle of a harmonic magnetic gear to transmit power from the second rotor (high-speed rotor) **40** to the first rotor (low-speed rotor) **30**.

(28) (Internal Structure of Magnetic Geared Electrical Machine)

(29) Subsequently, an internal structure of the above-described magnetic geared electrical machine **10** (**10A**, **10B**) will be described with reference to FIG. 2.

(30) FIG. 2 is a radial cross-sectional view of the magnetic geared electrical machine **10** according to an embodiment.

(31) As shown in FIG. 2, the stator **20** of the magnetic geared electrical machine **10** includes a plurality of stator magnets **22** and stator coils **24** arranged in the circumferential direction. The stator magnets **22** and the stator coils **24** are attached to a stator core **23**.

(32) The stator magnets **22** are composed of permanent magnets, and the plurality of stator magnets **22** are disposed in the circumferential direction so as to axially pass between the stator coils **24** and the first rotor **30** in the radial direction. In the example shown in FIG. 2, each of the stator magnets **22** is an axially elongated rod-shaped member having a rectangular cross section. That is, a dimension of each side of the rectangular cross section of the stator magnet **22** shown in FIG. 2 is sufficiently less than an axial dimension of the stator magnet **22** shown in FIGS. 1A and 1B.

(33) FIG. 2 shows a structural example of a surface permanent magnet (SPM) in which the stator magnets **22** are attached to the surface of the stator core **23**. In another embodiment, the stator **20** may have a structure of an interior permanent magnet (IPM) in which the stator magnets **22** are embedded in the stator core **23**.

(34) The stator coils **24** are disposed within a plurality of slots **25** disposed in the stator core **23**. The plurality of slots **25** are disposed in the circumferential direction, and each of the slots **25** extends in the axial direction. Both axial ends of each of the slots **25** are open, and coil ends of the stator coil **24** that do not fit into the slot **25** may protrude from the stator core **23** at both axial ends of the stator core **23**.

(35) The first rotor **30** opposed to the stator **20** of the above-described configuration in the radial direction includes a plurality of pole pieces **32** arranged in the circumferential direction with a first radial gap G1 between the stator **20** and the first rotor **30**. Each of the pole pieces **32** is composed of a magnetic material such as an electromagnetic steel plate or a powder magnetic core, and is an axially elongated rod-shaped member having the rectangular cross section (see FIG. 2). That is, a dimension of each side of the rectangular cross section of the pole piece **32** shown in FIG. 2 is sufficiently less than an axial dimension of the pole piece **32** shown in FIGS. 1A and 1B.

(36) Besides the pole pieces **32**, the first rotor **30** may include other members such as a non-magnetic member **33** (see FIG. 2) which is composed of a non-magnetic material and connects the pole pieces **32** in the circumferential direction and the first end plates **34** described above with reference to FIGS. 1A and 1B.

(37) The non-magnetic member **33** may be a fiber reinforced plastic (FRP) in which reinforcing fibers are combined with a matrix resin and may be, for example, CFRP using carbon fibers as reinforcing fibers or GFRP using glass fibers as reinforcing fibers. The first end plate **34** is an annular plate extending along the radial direction from a connection position with the rotational shaft **3**, **9** toward the pole piece **32**.

(38) As shown in FIG. 2, the second rotor **40** is disposed opposite to the stator **20** across the first rotor **30** of the above-described configuration in the radial direction with a second radial gap G2 between the first rotor **30** and the second rotor **40**. The second rotor **40** may be arranged to be

surrounded by the first rotor **30** including the pole piece **32**, the non-magnetic member **33**, and the first end plate **34** as described above with reference to FIGS. **1A** and **1B**.

(39) The first radial gap **G1** between the stator **20** and the first rotor **30** and the second radial gap **G2** between the first rotor **30** and the second rotor **40** may have substantially the same size, for example, $0.95 \leq G1/G2 \leq 1.05$.

(40) The second rotor **40** includes a plurality of rotor magnets **42** each of which is composed of a permanent magnet, and the plurality of rotor magnets **42** are arranged in the circumferential direction. Each of the rotor magnets **42** may be an axially elongated rod-shaped member having the rectangular cross section (see FIG. **2**). Herein, the rotor magnet **42** is “axially elongated” means that a dimension of each side of the rectangular cross section of the rotor magnet **42** shown in FIG. **2** is sufficiently less than an axial dimension of the rotor magnet **42** shown in FIGS. **1A** and **1B**.

(41) FIG. **2** shows a structural example of a surface permanent magnet (SPM) in which the rotor magnet **42** is attached to the surface of the rotor core **43**. In another embodiment, the second rotor **40** may have a structure of an interior permanent magnet (IPM) in which the rotor magnet **42** is embedded in the rotor core **43**.

(42) Further, in the exemplary embodiment shown in FIG. **2**, the plurality of rotor magnets **42** form at least two magnet groups (Gr1, Gr2) arranged with gaps in the circumferential direction on the surface of the rotor core **43**. For example, each of the magnet groups (Gr1, Gr2) is composed of the plurality of rotor magnets **42** having the same magnetization direction.

(43) Besides the rotor magnets **42** and the rotor core **43**, the second rotor **40** may include other members such as the second end plates **44** described above with reference to FIGS. **1A** and **1B** and a closure member **45** for filling the circumferential gap between the rotor magnets **42**.

(44) The second end plate **44** is an annular plate extending along the radial direction from a mounting position of the bearing **B2** toward the rotor core **43**.

(45) Further, the closure member **45** may be a fiber reinforced plastic (FRP) in which reinforcing fibers are combined with a matrix resin and may be, for example, CFRP using carbon fibers as reinforcing fibers or GFRP using glass fibers as reinforcing fibers. As shown in FIG. **2**, the closure member **45** may close at least part of the circumferential gap between the magnet groups (Gr1, Gr2) alternately arranged in the circumferential direction. In this case, a height from the rotor core **43** to a surface of the closure member **45** may be less than a protrusion height of each rotor magnet **42** from the rotor core **43**.

(46) In the exemplary embodiment shown in FIG. **2**, the number of stator coils **24**, rotor magnets **42**, pole pieces **32**, and stator magnets **22** are in ascending order.

(47) (Cooling Structure of Magnetic Geared Electrical Machine)

(48) In the magnetic geared electrical machine **10** (**10A**, **10B**) configured as described above, as a countermeasure against heat generation due to copper loss in the stator coil **24** or iron loss in the pole piece **32**, the magnetic geared electrical machine **10** is internally provided with a cooling gas flow path.

(49) FIG. **3** is a diagram showing the magnetic geared electrical machine **10** according to an embodiment. FIG. **4** is a diagram showing the magnetic geared electrical machine **10** according to another embodiment.

(50) In some embodiments, as shown in FIGS. **3** and **4**, the housing **21** which supports the stator **20** has a first inlet opening **102** for taking in a first cooling gas and a second inlet opening **202** for taking in a second cooling gas. The first cooling gas and the second cooling gas may be the same gas or different types of gases.

(51) The first inlet opening **102** communicates with a first inlet cavity **104** defined by at least the housing **21**. The first inlet cavity **104** communicates with a first radial gap **G1** between the stator **20** and the first rotor **30**.

(52) Similarly, the second inlet opening **202** communicates with a second inlet cavity **204** defined by at least the housing **21**. The second inlet cavity **204** communicates with a second radial gap **G2**

between the first rotor **30** and the second rotor **40**.

(53) In the examples shown in FIGS. **3** and **4**, the coil end formed by protruding the stator coil **24** from the stator core **23** in the axial direction faces the first inlet cavity **104**.

(54) Thus, two mutually independent cooling gas flows (i.e., a first cooling gas flow from the first inlet opening **102** through the first inlet cavity **104** and a second cooling gas flow from the second inlet opening **202** through the second inlet cavity **204**) are formed inside the magnetic geared electrical machine **10**.

(55) This facilitates cooling design according to the amount of heat generated or dissipated in each region on both sides of the pole pieces **32** of the magnetic geared electrical machine **10**.

(56) In the embodiments shown in FIGS. **3** and **4**, a partition member **70A** is disposed between the first inlet cavity **104** and the second inlet cavity **204**.

(57) By separating the inlet cavities (**104**, **204**) by the partition member **70A**, two mutually independent flows (i.e., a first flow through the first radial gap **G1** and a second flow through the second radial gap **G2**) can be easily achieved inside the magnetic geared electrical machine **10**.

(58) FIG. **5A** is a cross-sectional view of a partition member **70** according to an embodiment. FIG. **5B** is a cross-sectional view of the partition member **70** according to another embodiment. FIG. **5C** is a cross-sectional view of the partition member **70** according to another embodiment.

(59) As shown in FIGS. **5A** to **5C**, the partition member **70A** is fixed to at least one of the housing **21** or the first end plate **34** of the first rotor **30**.

(60) In some embodiments, the partition member **70A** includes a baffle plate **72** protruding from the housing **21** toward the first end plate **34** of the first rotor **30** or from the first end plate **34** of the first rotor **30** toward the housing **21**. In this case, a simple configuration of the baffle plate **72** can be used to prevent the flow of cooling gas between the inlet cavities (**104**, **204**) and increase the independence of the two types of flows inside the magnetic geared electrical machine **10**.

(61) In the exemplary embodiment shown in FIG. **5A**, the baffle plate **72** is disposed so as to protrude from the housing **21** along the axial direction toward the first end plate **34** of the first rotor **30**.

(62) In some embodiments, the partition member **70A** is a labyrinth seal **74**, **76** including a stationary fin **75A**, **77A** disposed on a side of the housing **21** and a rotating fin **75B**, **77B** disposed on a side of the first rotor **30**.

(63) In this case, the labyrinth seal **74**, **76** can be used to effectively prevent the flow of cooling gas between the inlet cavities (**104**, **204**) and increase the independence of the two types of flows inside the magnetic geared electrical machine **10**.

(64) In the labyrinth seal **74** shown in FIG. **5B**, the stationary fin **75A** protruding from the housing **21** along the axial direction toward the first end plate **34** and the rotating fin **75B** protruding from the first end plate **34** toward the housing **21** are arranged alternately in the radial direction.

(65) The labyrinth seal **76** shown in FIG. **5C** includes a support **78** protruding from the housing **21** along the axial direction toward the first end plate **34** and a support **79** protruding from the first end plate **34** toward the housing **21**, and the stationary fin **75A** and the rotating fin **77B** are provided on the supports **78**, **79**, respectively. Specifically, the stationary fin **77A** protrudes from the support **78** along the radial direction toward the support **79**. The rotating fin **77B**, which is arranged alternately with the stationary fin **77A** in the axial direction, protrudes from the support **79** along the radial direction toward the support **78**.

(66) As shown in FIGS. **3** and **4**, the first inlet cavity **104** may be defined by, together with the partition member **70A**, which may be the baffle plate **72** or the labyrinth seal **74**, **76**, a radial wall **21A** of the housing **21**, and an axial end face of the stator **20** and a first end plate **34A** of the first rotor **30**, which are opposed to the radial wall **21A**.

(67) Similarly, the second inlet cavity **204** may be defined by, together with the partition member **70A**, the radial wall **21A** of the housing **21** and the first end plate **34A** of the first rotor **30**, which is opposed to the radial wall **21A**.

(68) In some embodiments, as shown in FIGS. 3 and 4, the second inlet cavity 204 communicates with an upstream intermediate cavity 56 formed inside the first rotor 30 via an inlet opening 54 in one first end plate 34A.

(69) The upstream intermediate cavity 56 is an annular space formed between one first end plate 34A of the first rotor 30 and one second end plate 44A of the second rotor 40 opposed to the one first end plate 34A of the first rotor 30. An upstream end of the second radial gap G2 is fluidly connected to the upstream intermediate cavity 56, and the second cooling gas in the second inlet cavity 204 is introduced into the second radial gap G2 through the upstream intermediate cavity 56.

(70) In the embodiments shown in FIG. 3, the housing 21 has a first outlet opening 112 and a second outlet opening 212.

(71) The first outlet opening 112 communicates with a first outlet cavity 114 defined by at least the housing 21. The first outlet cavity 114 communicates with the first radial gap G1 between the stator 20 and the first rotor 30.

(72) Similarly, the second outlet opening 212 communicates with a second outlet cavity 214 defined by at least the housing 21. The second outlet cavity 214 communicates with the second radial gap G2 between the first rotor 30 and the second rotor 40.

(73) In the example shown in FIG. 3, the coil end formed by protruding the stator coil 24 from the stator core 23 in the axial direction faces the first outlet cavity 114.

(74) Thus, by providing separate paths inside the magnetic geared electrical machine 10 for the cooling gas having passed through each radial gap G1, G2 to be discharged from each outlet opening (112, 212), the degree of freedom of cooling design according to the amount of heat generated or dissipated in each region on both sides of the pole pieces 32 of the magnetic geared electrical machine 10 is further improved.

(75) In the embodiments shown in FIG. 3, a partition member 70B is disposed between the first outlet cavity 114 and the second outlet cavity 214.

(76) By separating the outlet cavities (114, 214) by the partition member 70B, two mutually independent flows (i.e., a first flow through the first radial gap G1 and a second flow through the second radial gap G2) can be easily achieved inside the magnetic geared electrical machine 10.

(77) The partition member 70B may be composed of the baffle plate 72 or the labyrinth seal 74, 76 described above with reference to FIGS. 5A to 5C, as well as the partition member 70A. As shown in FIG. 3, the first outlet cavity 114 may be defined by, together with the partition member 70B, which may be the baffle plate 72 or the labyrinth seal 74, 76, a radial wall 21B of the housing 21, and an axial end face of the stator 20 and a first end plate 34B of the first rotor 30, which are opposed to the radial wall 21B.

(78) Similarly, the second outlet cavity 214 may be defined by, together with the partition member 70B, the radial wall 21B of the housing 21 and the first end plate 34B of the first rotor 30, which is opposed to the radial wall 21B.

(79) As shown in FIG. 3, the second outlet cavity 214 communicates with a downstream intermediate cavity 58 formed inside the first rotor 30 via an outlet opening 74 in the other first end plate 34B.

(80) The downstream intermediate cavity 58 is an annular space formed between the other first end plate 34B of the first rotor 30 and the other second end plate 44B of the second rotor 40 opposed to the other first end plate 34B of the first rotor 30. A downstream end of the second radial gap G2 is fluidly connected to the downstream intermediate cavity 58, and the second cooling gas having passed through the second radial gap G2 is introduced into the second outlet cavity 214 through the downstream intermediate cavity 58.

(81) In some embodiments, as shown in FIG. 3, the magnetic geared electrical machine 10 includes a first circulation path 120 for returning the first cooling gas discharged from the first outlet opening 112 to the first inlet opening 102, and a first fan 122 disposed on the first circulation path 120. Further, the magnetic geared electrical machine 10 includes a second circulation path 220 for

returning the second cooling gas discharged from the second outlet opening **212** to the second inlet opening **202**, and a second fan **222** disposed on the second circulation path **220**.

(82) The first fan **122** and the second fan **222** are each controllable in capacity. Therefore, the flow rate of the first cooling gas through the first radial gap **G1** and the flow rate of the second cooling gas through the second radial gap **G2** can be adjusted independently of each other by capacity control of the first fan **122** and the second fan **222**. This allows appropriate cooling according to the amount of heat generated or dissipated in each region on both sides of the pole pieces **32** of the magnetic geared electrical machine **10**.

(83) In addition, as shown in FIG. **3**, the first circulation path **120** and the second circulation path **220** may be provided with heat exchangers **124**, **224** for cooling the hot first cooling gas and second cooling gas through heat exchange, respectively.

(84) In some embodiments, as shown in FIG. **4**, the housing **21** has a common outlet opening **312**. The common outlet opening **312** communicates with a common outlet cavity **314** defined by at least the housing **21**. The common outlet cavity **314** communicates with both the first radial gap **G1** between the stator **20** and the first rotor **30** and the second radial gap **G2** between the first rotor **30** and the second rotor **40**.

(85) By merging the cooling gas flow having passed through each radial gap **G1**, **G2** at the common outlet cavity **314** and discharging from the common outlet opening **312**, the flow path configuration inside the magnetic geared electrical machine **10** can be simplified.

(86) In the example shown in FIG. **4**, the coil end formed by protruding the stator coil **24** from the stator core **23** in the axial direction faces the common outlet cavity **314**.

(87) In the embodiment shown in FIG. **4**, the magnetic geared electrical machine **10** includes a circulation path **320** for returning the cooling gas discharged from the common outlet opening **312** to the first inlet opening **102** and the second inlet opening **202**, and a common fan **322** disposed on the circulation path **320**.

(88) Downstream of the common fan **322**, the circulation path **320** is divided into a first branch flow path **330** and a second branch flow path **332**, which are connected to the first inlet opening **102** and the second inlet opening **202**, respectively. Further, a first damper **340** is disposed on the first branch flow path **330**, and a second damper **342** is disposed on the second branch flow path **332**.

(89) The flow rate of the first cooling gas through the first radial gap **G1** and the flow rate of the second cooling gas through the second radial gap **G2** can be adjusted independently of each other by controlling the opening degree of the first damper **340** and the second damper **342**. This allows appropriate cooling according to the amount of heat generated or dissipated in each region on both sides of the pole pieces **32** of the magnetic geared electrical machine **10**.

(90) In addition, as shown in FIG. **4**, the circulation path **320** may be provided with a common heat exchanger **324** for cooling the hot cooling gas through heat exchange.

(91) In the embodiments shown in FIGS. **3** and **4**, the first cooling gas flowing through the first radial gap **G1** and the second cooling gas flowing through the second radial gap **G2** have the same flow direction, but a configuration in which the first and second cooling gases flow in opposite directions can also be employed.

(92) FIG. **6** is a diagram showing the magnetic geared electrical machine **10** according to an embodiment.

(93) As shown in this figure, the housing **21** has the first inlet opening **102** and the second outlet opening **212** on one end side in the axial direction, and has the first outlet opening **112** and the second inlet opening **202** on the other end side in the axial direction. Further, on one end side of the housing **21** in the axial direction, the first inlet cavity **104** and the second outlet cavity **214** adjacent to the first inlet cavity **104** in the radial direction are formed, with the first inlet opening **102** communicating with the first inlet cavity **104** and the second outlet opening **212** communicating with the second outlet cavity **214**. On the other hand, on the other end side of the housing **21** in the

axial direction, the first outlet cavity **114** and the second inlet cavity **204** adjacent to the first outlet cavity **114** in the radial direction are formed, with the first outlet opening **112** communicating with the first outlet cavity **114** and the second inlet opening **202** communicating with the second inlet cavity **204**. The flow direction of the first cooling gas from the first inlet cavity **104** through the first radial gap **G1** to the first outlet cavity **114** is opposite to the flow direction of the second cooling gas from the second inlet cavity **204** through the second radial gap **G2** to the second outlet cavity **214**. By flowing the first cooling gas and the second cooling gas in opposite directions, the temperature rise of the pole pieces **32** can be appropriately suppressed regardless of the axial position.

(94) In the exemplary embodiment shown in FIG. 6, a partition member **70C** is disposed between the first inlet cavity **104** and the second outlet cavity **214**, and a partition member **70D** is disposed between the first outlet cavity **114** and the second inlet cavity **204**. The partition members **70C**, **70D** may be composed of the baffle plate **72** or the labyrinth seal **74**, **76** described above with reference to FIGS. 5A to 5C, as well as the partition member **70A**.

(95) (Tandem Arrangement of Magnetic Geared Units)

(96) In the embodiments shown in FIGS. 3, 4, and 6, the magnetic geared electrical machine **10** includes one set of the stator **20**, the first rotor **30**, and the second rotor **40**, but the magnetic geared electrical machine **10** may include a plurality of magnetic geared units each of which is composed of the stator **20**, the first rotor **30**, and the second rotor **40**.

(97) FIG. 7 is a diagram showing the magnetic geared electrical machine **10** according to an embodiment.

(98) As shown in this figure, the magnetic geared electrical machine **10** includes a plurality of magnetic geared units **400** (**400A**, **400B**) each of which is composed of the stator **20**, the first rotor **30**, and the second rotor **40**. The plurality of magnetic geared units **400** (**400A**, **400B**) are arranged in the axial direction. The respective first radial gaps **G1** of the magnetic geared units **400** (**400A**, **400B**) are connected fluidly in parallel to each other so that the first cooling gas passes through the respective first radial gaps **G1** of the magnetic geared units **400** (**400A**, **400B**) in opposite directions.

(99) In the example shown in FIG. 7, the first inlet opening **102** and the first inlet cavity **104** are disposed between a pair of magnetic geared units **400A**, **400B** which are adjacent in the axial direction. In this case, the first cooling gas flowing into the first inlet cavity **104** is diverted to both axial sides and passes through the respective first radial gaps **G1** of the magnetic geared units **400** (**400A**, **400B**) in opposite directions to each other. The flow path of the first cooling gas formed between the pair of axially adjacent magnetic geared units **400A**, **400B** may be a flow path inlet portion of the first cooling gas (first inlet opening **102** and first inlet cavity **104**) as in the example in FIG. 7, or may be a flow path outlet portion of the first cooling gas (first outlet opening **112** and first outlet cavity **114**). That is, in another embodiment, the first outlet opening **112** and the first outlet cavity **114** are disposed between the pair of magnetic geared units **400A**, **400B** which are adjacent in the axial direction. In this case, the first cooling gas having passed through the respective first radial gaps **G1** of the magnetic geared units **400** (**400A**, **400B**) in opposite directions flows into the first outlet cavity **114** between the magnetic geared units **400A**, **400B** and is discharged from the first outlet opening **112** between the magnetic geared units **400A**, **400B**.

(100) In the example shown in FIG. 7, the first inlet openings **102** are provided at two axial positions for the two magnetic geared units **400A**, **400B**, but the first inlet opening **102** may be provided at one axial position for the two magnetic geared units **400A**, **400B**. Similarly, in the other embodiment in which the first outlet opening **112** and the first outlet cavity **114** are disposed between the pair of axially adjacent magnetic geared units **400A**, **400B**, the first outlet opening(s) **112** may be provided at one or more axial positions for the two magnetic geared units **400A**, **400B**.

(101) The stator **20** side of the magnetic geared electrical machine **10** with respect to the pole pieces **32** is affected by the heat generated by the stator coil **24** in addition to the heat generated by

the pole pieces **32** themselves, and depending on the specifications of the magnetic geared electrical machine **10**, relatively high cooling performance is required. When the plurality of magnetic geared units **400** are arranged in the axial direction, if a flow path configuration is adopted such that the first cooling gas sequentially passes through the first radial gaps **G1** of the plurality of magnetic geared units **400**, the temperature of the first cooling gas may rise in the first radial gap **G1** of the most downstream magnetic geared unit **400**, resulting in insufficient cooling performance.

(102) In this regard, by adopting a flow path configuration in which the first radial gaps **G1** of the magnetic geared units **400** (**400A**, **400B**) are connected fluidly in parallel to each other as described above, compared to the flow path configuration in which the first cooling gas sequentially passes through the first radial gaps **G1** of the plurality of magnetic geared units **400**, the supply temperature of the first cooling gas can be reduced. As a result, it is possible to improve the cooling performance on the stator **20** side of the magnetic geared electrical machine **10** with respect to the pole pieces **32**.

(103) On the other hand, on the second rotor **40** side of the magnetic geared electrical machine **10** with respect to the pole pieces **32**, not so high cooling performance may not be required.

(104) For this reason, in the embodiment shown in FIG. 7, the second inlet opening **202** and the second inlet cavity **204** are disposed at one end of the magnetic geared electrical machine **10** in the axial direction so that the second cooling gas taken into the second inlet cavity **204** sequentially passes through the second radial gaps **G2** of the plurality of magnetic geared units **400** (**400A**, **400B**).

(105) This allows the second inlet opening **202** and the second inlet cavity **204** to be placed at one end of the magnetic geared electrical machine **10** in the axial direction, simplifying the connection structure of the piping for introducing the second cooling gas from the outside.

(106) When the second cooling gas sequentially passes through the second radial gaps **G2** of the plurality of magnetic geared units **400** (**400A**, **400B**), the temperature of the second cooling gas may rise at the outlet side of the second radial gap **G2** of the downstream magnetic geared unit **400B**, resulting in insufficient cooling of the pole pieces **32**. In this case, by adopting a flow path configuration in which the first cooling gas having passed through the respective first radial gaps **G1** flows into the first outlet cavity **114** between the axially adjacent magnetic geared units **400A**, **400B**, the first cooling gas and the second cooling gas flow in opposite directions in the magnetic geared unit **400B**. Thus, it is possible to appropriately suppress the temperature rise of the pole pieces **32** in the magnetic geared unit **400B**, in which the temperature rise of the second cooling gas may pose a problem.

(107) Further, in some embodiments, the magnetic geared electrical machine **10** includes a partition wall **420** connecting the first end plates **34** of the first rotors **30** of the pair of axially adjacent magnetic geared units **400A**, **400B** so as to separate a flow path of the first cooling gas from a flow path (central cavity **410**) of the second cooling gas.

(108) In the exemplary embodiment shown in FIG. 7, the flow path of the first cooling gas separated from the central cavity **410** by the partition wall **420** is the first inlet cavity **104** disposed between the pair of axially adjacent magnetic geared units **400A**, **400B**. In another embodiment, the flow path of the first cooling gas separated from the central cavity **410** by the partition wall **420** is the first outlet cavity **114** disposed between the pair of axially adjacent magnetic geared units **400A**, **400B**.

(109) Since the first rotors **30** of the pair of axially adjacent magnetic geared units **400A**, **400B** have the same rotation speed, the flow path of the first cooling gas (first inlet cavity **104** or first outlet cavity **114**) and the flow path of the second cooling gas (central cavity **410**) can be easily separated by a simple structure of the partition wall **420** connecting the first rotors **30**.

(110) (Additional Modification for Partition Member)

(111) FIGS. **8A** and **8B** are diagrams showing an arrangement example of the partition member **70**.

(112) Consider the case where a second cavity (204, 214) which is either one of the second inlet cavity 204 or the second outlet cavity 214 is disposed radially inward of a first cavity (104, 114) which is either one of the first inlet cavity 104 or the first outlet cavity 114. In this case, as shown in FIG. 8A, if the partition member 70 separating the first cavity (104, 114) from the second cavity (204, 214) is disposed near the first radial gap G1 with respect to the radial direction, due to centrifugal force, the second cooling gas forms vortices and is retained in the region radially outward of the inlet opening 54 and the outlet opening 74 of the second cavity (204, 214).

(113) In contrast, in the example shown in FIG. 8B, the partition member 70 separating the first cavity (104, 114) from the second cavity (204, 214) is disposed closer to the inlet opening 54 and the outlet opening 74 than the first radial gap G1. In other words, the partition member 70 is disposed radially inward of the midpoint of the distance L in the radial direction between the first radial gap G1 and the inlet opening 54 or the outlet opening 74. This suppresses the retention of the second cooling gas in the second cavity (204, 214). In the first cavity (104, 114), vortices are formed radially inward of the first radial gap G1, but they flow downstream while accompanying the main flow of the first cooling gas flowing through the first inlet cavity 104, the first radial gap G1, and the first outlet cavity 114 and discharged through the first outlet opening 112, so they are difficult to stay.

(114) FIG. 9 is a diagram showing an arrangement example of the partition member 70 according to another embodiment.

(115) As shown in this figure, in some embodiments, a distance d between the partition member 70 separating the first cavity (104, 114) from the second cavity (204, 214) and the opening (54, 74) of the first end plate 34 is less than a radial dimension D of the opening (54, 74).

(116) Since the radial distance d between the partition member 70 and the opening (54, 74) of the end plate (34A, 34B) is set to a relatively small value, the retention of the second cooling gas in the second cavity (204, 214) is suppressed.

(117) (Conclusion)

(118) Hereinafter, the overview of the magnetic geared electrical machine 10 and the power generation system 1 using the same according to some embodiments will be described.

(119) (1) A magnetic geared electrical machine (10) according to some embodiments of the present invention includes: a stator (20) including a stator coil (24); a first rotor (30) including a plurality of pole pieces (32); a second rotor (40) including a plurality of rotor magnets (42), and arranged opposite to the stator (20) across the first rotor (30) in a radial direction; a housing (21) having a first inlet opening (102) and a second inlet opening (202), and supporting the stator (20); a first inlet cavity (104) defined by at least the housing (21) so as to communicate with the first inlet opening (102) provided in the housing (21) and with a first radial gap (G1) between the stator (20) and the first rotor (30); and a second inlet cavity (204) defined by at least the housing (21) so as to communicate with the second inlet opening (202) provided in the housing (21) and with a second radial gap (G2) between the first rotor (30) and the second rotor (40).

(120) In the above configuration (1), two types of inlet cavities (first inlet cavity (104) and second inlet cavity (204)) respectively communicating with two types of inlet openings (first inlet opening (102) and second inlet opening (202)) in the housing (21) which supports the stator (20) are formed inside the housing (21).

(121) Thereby, two mutually independent cooling gas flows, i.e., a first flow of cooling gas through the first radial gap (G1) and a second flow of cooling gas through the second radial gap (G2), can be formed inside the magnetic geared electrical machine (10). Herein, “mutually independent” means that the cooling gas flows in substantially separate paths from the inlet to each radial gap. Specifically, the first flow is the flow of cooling gas from the first inlet opening (102) through the first inlet cavity (104) to the first radial gap (G1). In contrast, the second flow is the flow of cooling gas from the second inlet opening (202) through the second inlet cavity (204) to the second radial gap (G2). These two types of cooling gas flows are “mutually independent” because they are

substantially separate paths from the inlet opening to each radial gap.

(122) Thus, by forming two mutually independent cooling gas flows inside the magnetic geared electrical machine (10), it is possible to easily make cooling design according to the amount of heat generated or dissipated in each region on both sides of the pole pieces (32) of the magnetic geared electrical machine (10). In contrast, for example, when the cooling gas taken in from a common inlet opening is divided in the middle and passes through the first radial gap (G1) and the second radial gap (G2), the flow distribution ratio of the cooling gas between the first radial gap (G1) and the second radial gap (G2) depends on the size of the flow path cross sections of the two gaps, so it is difficult to design a cooling system with a high degree of freedom according to the amount of heat generated or dissipated in each region on both sides of the pole pieces (32) of the magnetic geared electrical machine (10).

(123) (2) In some embodiments, in the above configuration (1), the magnetic geared electrical machine (10) includes a partition member (70) disposed between the first inlet cavity (104) and the second inlet cavity (204).

(124) With the above configuration (2), by separating the inlet cavities by the partition member (70), two mutually independent flows (i.e., a first flow of cooling gas through the first radial gap (G1) and a second flow of cooling gas through the second radial gap (G2)) can be easily achieved inside the magnetic geared electrical machine (10).

(125) (3) In some embodiments, in the above configuration (2), the partition member (70) includes a labyrinth seal (74, 76) which is formed by a stationary fin disposed on a side of the housing (21) and a rotating fin disposed on a side of the first rotor (30).

(126) With the above configuration (3), the labyrinth seal (74, 76) can be used to effectively prevent the flow of cooling gas between the inlet cavities and increase the independence of the two types of flows inside the magnetic geared electrical machine (10).

(127) (4) In some embodiments, in the above configuration (2), the partition member (70) includes a baffle plate (72) protruding from the housing (21) toward the first rotor (30) or from the first rotor (30) toward the housing (21).

(128) With the above configuration (4), a simple configuration of the baffle plate (72) can be used to prevent the flow of cooling gas between the inlet cavities and increase the independence of the two types of flows inside the magnetic geared electrical machine (10).

(129) (5) In some embodiments, in any one of the above configurations (1) to (4), the magnetic gear generator (10) includes: a plurality of magnetic geared units (400) arranged in an axial direction, each of the magnetic geared units (400) being composed of the stator (20), the first rotor (30), and the second rotor (40). The respective first radial gaps (G1) of the magnetic geared units (400) are connected fluidly in parallel to each other so that a first cooling gas passes through the respective first radial gaps (G1) of the magnetic geared units (400) in opposite directions.

(130) The stator (20) side of the magnetic geared electrical machine (10) with respect to the pole pieces (32) is affected by the heat generated by the stator coil (24) in addition to the heat generated by the pole pieces (32) themselves, and depending on the specifications of the magnetic geared electrical machine (10), relatively high cooling performance is required. When the plurality of magnetic geared units are arranged in the axial direction, if a flow path configuration is adopted such that the first cooling gas sequentially passes through the first radial gaps (G1) of the plurality of magnetic geared units (i.e., a flow path configuration in which the first radial gaps G1 are connected in series), the temperature of the first cooling gas may rise in the first radial gap (G1) of the most downstream magnetic geared unit, resulting in insufficient cooling performance.

(131) In this regard, with the above configuration (5), by adopting a flow path configuration in which the first radial gaps (G1) of the magnetic geared units are connected in parallel to each other, compared to the flow path configuration in which the first cooling gas sequentially passes through the first radial gaps (G1), the supply temperature of the first cooling gas can be reduced. As a result, it is possible to improve the cooling performance on the stator (20) side of the magnetic

geared electrical machine (10) with respect to the pole pieces (32).

(132) (6) In some embodiments, in the above configuration (5), the second inlet opening (202) and the second inlet cavity (204) are disposed at one end of the magnetic geared electrical machine (10) in the axial direction so that a second cooling gas taken into the second inlet cavity (204) sequentially passes through the second radial gaps (G2) of the plurality of magnetic geared units (400).

(133) With the above configuration (6), the second inlet opening (202) and the second inlet cavity (204) can be placed at one end of the magnetic geared electrical machine (10) in the axial direction, simplifying the connection structure of the piping for introducing the second cooling gas from the outside.

(134) (7) In some embodiments, in the above configuration (6), the magnetic gear generator (10) includes a partition wall (420) connecting the first rotors (30) of a pair of magnetic geared units (400) which are adjacent in the axial direction so as to separate a flow path of the first cooling gas from a flow path of the second cooling gas.

(135) With the above configuration (7), since the first rotors (30) of the pair of axially adjacent magnetic geared units (400) have the same rotation speed, the flow path of the first cooling gas and the flow path of the second cooling gas can be easily separated by a simple structure of the partition wall (420) connecting the first rotors (30).

(136) (8) In some embodiments, in any one of the above configurations (1) to (7), the housing (21) has a first outlet opening (112) and a second outlet opening (212). The magnetic geared electrical machine (10) includes: a first outlet cavity (112) defined by at least the housing (21) so as to communicate with the first outlet opening (112) provided in the housing (21) and with the first radial gap (G1); and a second outlet cavity (214) defined by at least the housing (21) so as to communicate with the second outlet opening (212) provided in the housing (21) and with the second radial gap (G2).

(137) With the above configuration (8), by providing separate paths in the magnetic geared electrical machine (10) for the cooling gas having passed through each radial gap to be discharged from each outlet opening, the degree of freedom of cooling design according to the amount of heat generated or dissipated in each region on both sides of the pole pieces (32) of the magnetic geared electrical machine (10) is further improved.

(138) For example, the flow rate of the first cooling gas through the first radial gap (G1) and the flow rate of the second cooling gas through the second radial gap (G2) may be adjusted independently of each other by adjusting design parameters such as the presence or absence of contraction and the opening area of contraction for each path until the cooling gas after passing through each radial gap is discharged from each outlet opening.

(139) (9) In some embodiments, in the above configuration (8), the magnetic gear generator (10) includes a plurality of magnetic geared units (400) arranged in an axial direction, each of the magnetic geared units (400) being composed of the stator (20), the first rotor (30), and the second rotor (40). The first outlet opening (112) and the first outlet cavity (114) are disposed between a pair of magnetic geared units (400) which are adjacent in the axial direction so that a first cooling gas that has passed through the respective first radial gaps (G1) of the pair of magnetic geared units (400) in opposite directions flows into the first outlet cavity (114).

(140) When the second cooling gas sequentially passes through the second radial gaps (G2) of the plurality of magnetic geared units 400 (400A, 400B), the temperature of the second cooling gas may rise at the outlet side of the second radial gap (G2) of the downstream magnetic geared unit (400B), resulting in insufficient cooling.

(141) In this regard, as in the above (9), by adopting a flow path configuration in which the first cooling gas having passed through the respective first radial gaps (G1) flows into the first outlet cavity (114) between the axially adjacent magnetic geared units (400A, 400B), the first cooling gas and the second cooling gas flow in opposite directions in the magnetic geared unit (400B) on the

downstream side with respect to the flow of the second cooling gas. Thus, it is possible to appropriately suppress the temperature rise of the pole pieces (32) in the downstream magnetic geared unit (400B), in which the temperature rise of the second cooling gas may pose a problem. (142) (10) In some embodiments, in the above configuration (8) or (9), the magnetic geared electrical machine (10) includes: a first circulation path for returning a first cooling gas discharged from the first outlet opening (112) to the first inlet opening (102); a second circulation path for returning a second cooling gas discharged from the second outlet opening (212) to the second inlet opening (202); a first fan with controllable capacity disposed on the first circulation path; and a second fan with controllable capacity disposed on the second circulation path.

(143) With the above configuration (10), by independently controlling the capacity of the first and second fans, the flow rate of the first cooling gas passing through the first radial gap (G1) and the flow rate of the second cooling gas passing through the second radial gap (G2) can be adjusted independently of each other. This allows appropriate cooling according to the amount of heat generated or dissipated in each region on both sides of the pole pieces (32) of the magnetic geared electrical machine (10).

(144) (11) In some embodiments, in any one of the above configurations (8) to (10), the housing (21) has the first inlet opening (102) and the second outlet opening (212) on one end side in an axial direction with respect to the stator (20), the first rotor (30), and the second rotor (40), and has the first outlet opening (112) and the second inlet opening (202) on another end side in the axial direction.

(145) With the above configuration (11), the flow direction of the first cooling gas in the first radial gap (G1) and the flow direction of the second cooling gas in the second radial gap (G2) are opposite (counter flow), and the temperature rise of the pole pieces (32) can be appropriately suppressed regardless of the axial position.

(146) (12) In some embodiments, in any one of the above configurations (8) to (11), the stator (20) is disposed radially outward of the second rotor (40) with the first rotor (30) interposed therebetween. The first rotor (30) includes a pair of end plates disposed at both axial ends of the plurality of pole pieces (32) and having an opening communicating with the second radial gap (G2). The magnetic geared electrical machine comprises a partition member (70) disposed between a first cavity which is either one of the first inlet cavity (104) or the first outlet cavity (114) and a second cavity which is the second inlet cavity (204) or the second outlet cavity (214) that is disposed radially inward of the first cavity. The partition member (70) is disposed radially inward of a center position in the radial direction between the first radial gap (G1) and the opening.

(147) With the above configuration (12), since the partition member (70) is disposed radially inward of the center position between the first radial gap (G1) and the opening in the end plate to separate the first cavity from the second cavity, the retention of the second cooling gas in the second cavity is suppressed.

(148) (13) In some embodiments, in any one of the above configurations (8) to (12), the stator (20) is disposed radially outward of the second rotor (40) with the first rotor (30) interposed therebetween. The first rotor (30) includes a pair of end plates disposed at both axial ends of the plurality of pole pieces (32) and having an opening communicating with the second radial gap (G2). The magnetic geared electrical machine comprises a partition member (70) disposed between a first cavity which is either one of the first inlet cavity (104) or the first outlet cavity (114) and a second cavity which is the second inlet cavity (204) or the second outlet cavity (214) that is disposed radially inward of the first cavity. A distance in the radial direction between the partition member (70) and the opening is less than a radial dimension of the opening.

(149) With the above configuration (13), since the radial distance between the partition member (70) separating the first cavity from the second cavity and the opening of the end plate is set to a relatively small value, the retention of the second cooling gas in the second cavity is suppressed.

(150) (14) In some embodiments, in any one of the above configurations (1) to (4), the housing

(21) has a common outlet opening (312). The magnetic geared electrical machine (10) includes a common outlet cavity (314) defined by at least the housing (21) so as to communicate with the common outlet opening (312) provided in the housing (21) and with both the first radial gap (G1) and the second radial gap (G2).

(151) With the above configuration (14), by merging the cooling gas flow having passed through each radial gap at the common outlet cavity and discharging from the common outlet opening (312), the flow path configuration inside the magnetic geared electrical machine (10) can be simplified. It is also advantageous in that it reduces the number of cooling gas outlet pipes connected to the common outlet opening (312) and enables a compact outlet piping layout.

(152) (15) A power generation system (1) according to some embodiments of the present disclosure includes: a prime mover (2); and a magnetic geared generator (10) driven by the prime mover (2). The magnetic geared generator is configured by the magnetic geared electrical machine (10) described in any one of the above (1) to (14).

(153) With the above configuration (15), as described in the above (1), since two types of inlet cavities (first inlet cavity (104) and second inlet cavity (204)) respectively communicating with two types of inlet openings (first inlet opening (102) and second inlet opening (202)) in the housing (21) which supports the stator (20) are formed inside the housing (21), it is possible to easily make cooling design according to the amount of heat generated or dissipated in each region on both sides of the pole pieces (32) of the magnetic geared electrical machine (10).

(154) Embodiments of the present invention were described in detail above, but the present invention is not limited thereto, and various amendments and modifications may be implemented.

(155) For example, the magnetic geared electrical machine 10 may include a cooling jacket for cooling the stator 20. The cooling jacket may be disposed between the stator core 23 and the housing 21 in the radial direction, or may be disposed inside the stator core 23. The cooling jacket is supplied with a cooling medium (for example, cooling water or cooling air) via a cooling medium supply pipe (not shown) connected to the cooling jacket. The cooling medium increased in temperature by taking heat from the stator 20 is discharged from a cooling medium discharge pipe (not shown) connected to the cooling jacket.

(156) In the present specification, an expression of relative or absolute arrangement such as “in a direction”, “along a direction”, “parallel”, “orthogonal”, “centered”, “concentric” and “coaxial” shall not be construed as indicating only the arrangement in a strict literal sense, but also includes a state where the arrangement is relatively displaced by a tolerance, or by an angle or a distance whereby it is possible to achieve the same function.

(157) For instance, an expression of an equal state such as “same” “equal” and “uniform” shall not be construed as indicating only the state in which the feature is strictly equal, but also includes a state in which there is a tolerance or a difference that can still achieve the same function.

(158) Further, an expression of a shape such as a rectangular shape or a cylindrical shape shall not be construed as only the geometrically strict shape, but also includes a shape with unevenness or chamfered corners within the range in which the same effect can be achieved.

(159) On the other hand, an expression such as “comprise”, “include”, and “have” are not intended to be exclusive of other components.

REFERENCE SIGNS LIST

(160) 1 Power generation system 2 Prime mover 3, 9 Rotational shaft 4 Power supply destination 6 Power supply source 8 Rotating machine 10 Magnetic geared electrical machine 10A Magnetic geared generator 10B Magnetic geared motor 20 Stator 21 Housing 21A, 21B Radial wall 22 Stator magnet 23 Stator core 24 Stator coil 25 Slot 30 First rotor 32 Pole piece 33 Non-magnetic member 34, 34A, 34B First end plate 40 Second rotor 42 Rotor magnet 43 Rotor core 44, 44A, 44B Second end plate 45 Closure member 54, 74, 212 Opening 56 Upstream intermediate cavity 58 Downstream intermediate cavity 70, 70A, 70B, 70C, 70D Partition member 72 Baffle plate 74 Labyrinth seal 74, 76 Labyrinth seal 75A, 77A Stationary fin 75B, 77B Rotating fin 78, 79 Support

102 First inlet opening **104** First inlet cavity **112** First outlet opening **114** First outlet cavity **120** First circulation path **122** First fan **124**, **224** Heat exchanger **202** Second inlet opening **204** Second inlet cavity **212** Second outlet opening **214** Second outlet cavity **220** Second circulation path **222** Second fan **312** Common outlet opening **314** Common outlet cavity **320** Circulation path **322** Common fan **324** Common heat exchanger **330** First branch flow path **332** Second branch flow path **340** First damper **342** Second damper **400**, **400A**, **400B** Magnetic geared unit **410** Central cavity **420** Partition wall **B1**, **B2** Bearing **G1** First radial gap **G2** Second radial gap

Claims

1. A magnetic geared electrical machine, comprising: a stator including a stator coil; a first rotor including a plurality of pole pieces; a second rotor including a plurality of rotor magnets, and arranged opposite to the stator across the first rotor in a radial direction; a housing having a first inlet opening and a second inlet opening, and supporting the stator; a first inlet cavity defined by at least the housing so as to communicate with the first inlet opening provided in the housing and with a first radial gap between the stator and the first rotor; and a second inlet cavity defined by at least the housing so as to communicate with the second inlet opening provided in the housing and with a second radial gap between the first rotor and the second rotor.
2. The magnetic geared electrical machine according to claim 1, further comprising a partition member disposed between the first inlet cavity and the second inlet cavity.
3. The magnetic geared electrical machine according to claim 2, wherein the partition member includes a labyrinth seal which is formed by a stationary fin disposed on a side of the housing and a rotating fin disposed on a side of the first rotor.
4. The magnetic geared electrical machine according to claim 2, wherein the partition member includes a baffle plate protruding from the housing toward the first rotor or from the first rotor toward the housing.
5. The magnetic geared electrical machine according to claim 1, further comprising: a plurality of magnetic geared units arranged in an axial direction, each of the magnetic geared units being composed of the stator, the first rotor, and the second rotor, wherein the respective first radial gaps of the magnetic geared units are connected fluidly in parallel to each other so that a first cooling gas passes through the respective first radial gaps of the magnetic geared units in opposite directions.
6. The magnetic geared electrical machine according to claim 5, wherein the second inlet opening and the second inlet cavity are disposed at one end of the magnetic geared electrical machine in the axial direction so that a second cooling gas taken into the second inlet cavity sequentially passes through the second radial gaps of the plurality of magnetic geared units.
7. The magnetic geared electrical machine according to claim 6, further comprising a partition wall connecting the first rotors of a pair of magnetic geared units which are adjacent in the axial direction so as to separate a flow path of the first cooling gas from a flow path of the second cooling gas.
8. The magnetic geared electrical machine according to claim 1, wherein the housing has a first outlet opening and a second outlet opening, and the magnetic geared electrical machine comprises: a first outlet cavity defined by at least the housing so as to communicate with the first outlet opening provided in the housing and with the first radial gap; and a second outlet cavity defined by at least the housing so as to communicate with the second outlet opening provided in the housing and with the second radial gap.
9. The magnetic geared electrical machine according to claim 8, further comprising: a plurality of magnetic geared units arranged in an axial direction, each of the magnetic geared units being composed of the stator, the first rotor, and the second rotor, wherein the first outlet opening and the first outlet cavity are disposed between a pair of magnetic geared units which are adjacent in the

axial direction so that a first cooling gas that has passed through the respective first radial gaps of the pair of magnetic geared units in opposite directions flows into the first outlet cavity.

10. The magnetic geared electrical machine according to claim 8, further comprising: a first circulation path for returning a first cooling gas discharged from the first outlet opening to the first inlet opening; a second circulation path for returning a second cooling gas discharged from the second outlet opening to the second inlet opening; a first fan with controllable capacity disposed on the first circulation path; and a second fan with controllable capacity disposed on the second circulation path.

11. The magnetic geared electrical machine according to claim 8, wherein the housing has the first inlet opening and the second outlet opening on one end side in an axial direction with respect to the stator, the first rotor, and the second rotor, and has the first outlet opening and the second inlet opening on another end side in the axial direction.

12. The magnetic geared electrical machine according to claim 8, wherein the stator is disposed radially outward of the second rotor with the first rotor interposed therebetween, the first rotor includes a pair of end plates disposed at both axial ends of the plurality of pole pieces and having an opening communicating with the second radial gap, the magnetic geared electrical machine comprises a partition member disposed between a first cavity which is either one of the first inlet cavity or the first outlet cavity and a second cavity which is the second inlet cavity or the second outlet cavity that is disposed radially inward of the first cavity, and the partition member is disposed radially inward of a center position in the radial direction between the first radial gap and the opening.

13. The magnetic geared electrical machine according to claim 8, wherein the stator is disposed radially outward of the second rotor with the first rotor interposed therebetween, the first rotor includes a pair of end plates disposed at both axial ends of the plurality of pole pieces and having an opening communicating with the second radial gap, the magnetic geared electrical machine comprises a partition member disposed between a first cavity which is either one of the first inlet cavity or the first outlet cavity and a second cavity which is the second inlet cavity or the second outlet cavity that is disposed radially inward of the first cavity, and a distance in the radial direction between the partition member and the opening is less than a radial dimension of the opening.

14. The magnetic geared electrical machine according to claim 1, wherein the housing has a common outlet opening, and the magnetic geared electrical machine comprises a common outlet cavity defined by at least the housing so as to communicate with the common outlet opening provided in the housing and with both the first radial gap and the second radial gap.

15. A power generation system, comprising: a prime mover; and a magnetic geared generator driven by the prime mover, wherein the magnetic geared generator is configured by the magnetic geared electrical machine according to claim 1.
